SGA 6, Exposé VI

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2.8. End of the proof of Theorem 2.1

The proof is finished by induction. Indeed, if $0 \leq s \leq r$ and if $x_j = 0$ for $s < j \leq r$, one has

$$\sum_{i=0}^{s} f^*(x_i)\xi^i = 0,$$

hence

$$\sum_{i=0}^{s} f^*(x_i)\xi^{i+r-s} = 0,$$

and consequently $x_s = 0$, by what precedes.

3. Consequences of the structure theorem for the $K^\bullet$ of a projective bundle

Proposition 3.1. Let S be a quasi-compact scheme, let $E_1, \dots, E_n$ be locally free $\mathcal{O}_S$-modules of finite type, let $D(E_i)$ be the flag scheme of E_i, and put

$$X = D(E_1) \times_S \cdots \times_S D(E_n).$$

Then the canonical homomorphism

$$K^\bullet(S) \longrightarrow K^\bullet(X)$$

is injective, whether $K^\bullet$ is defined from locally free $\mathcal{O}_S$-modules of finite type or from perfect complexes.

Using the universal property of flag schemes, one immediately checks that X may be constructed as follows. Consider the sequence of S-schemes $X_1, \dots, X_n$ defined recursively by taking X_k to be the flag scheme of the inverse image of E_k on X_{k-1}; then $X = X_n$. Thus, by induction, it is enough to treat the case $n = 1$, namely the case in which X is the flag scheme of a locally free $\mathcal{O}_S$-module E of finite type.

In that case X may in turn be constructed by successive projective bundles. Put $E = F_0$, $S = X_0$, and define $X_1, \dots, X_n$ and, on each X_i, a locally free module F_i by

$$X_k = \mathbf{P}(F_{k-1}), \qquad F_k = \mathrm{Ker}\big(f_k^*(F_{k-1}) \longrightarrow \mathcal{O}_{X_k}(1)\big),$$

where $f_k : X_k \to X_{k-1}$ is the structural morphism. It is therefore enough to prove the assertion when X is a projective bundle over S; this follows immediately from 1.1, respectively 2.1.

Theorem 3.2. Let S be a quasi-compact scheme. The pre-λ-ring $K^\bullet(S)$, defined from the locally free $\mathcal{O}_S$-modules of finite type, is a λ-ring.

Since the homomorphism

$$\lambda_t : K^\bullet(S) \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

is additive, to verify that it is a λ-homomorphism it is enough to verify the universal relations on elements of the form $\mathrm{cl}(E)$, where E is a locally free $\mathcal{O}_S$-module of finite type. Thus one is reduced to showing that for every pair of elements $E, F \in K^\bullet(S)$, classes of locally free finite-type $\mathcal{O}_S$-modules E, F, the relations V 2.4.2 are satisfied:

$$\lambda^i(EF) = P_i(\lambda^1(E), \dots, \lambda^i(E); \lambda^1(F), \dots, \lambda^i(F)),$$

$$\lambda^j(\lambda^i(E)) = Q_{ij}(\lambda^1(E), \dots, \lambda^{ij}(E)).$$

Since S is quasi-compact, there is a finite covering of S by disjoint open subsets on each of which E and F have constant rank. Since $K^\bullet$ of a finite disjoint union of opens is the product of their $K^\bullet$'s, one may assume that E and F have constant rank on S.

Let

$$X = D(E) \times_S D(F).$$

The homomorphism $K^\bullet(S) \to K^\bullet(X)$ is injective by 3.1. Hence, to prove the relations V 2.4.2, one may work in $K^\bullet(X)$. There the elements E and F split as sums of classes of invertible sheaves. By additivity, it is enough to verify the same relations for elements ξ, η, classes of invertible sheaves. Written in the form V 2.4.1, they become

$$1 + \xi\eta t = (1 + \xi t) \circ (1 + \eta t), \qquad \lambda_u(1 + \xi t) = 1 + \{1 + \xi t\}u.$$

These are precisely the relations V 2.3.1 and 2.3.3 defining the λ-ring structure of $1 + K^\bullet(S)[[t]]^+$.

Theorem 3.3. Let (T, A) be a commutatively ringed topos, and let K denote the Grothendieck group of A-modules locally free of bounded rank on T. Then K, endowed with its canonical pre-λ-structure (V 2.2 b)), is a λ-ring.

To prove that K is a λ-ring, one has to prove the relations V 2.4.1; by additivity, it is enough to prove them for elements E, F which are classes of A-modules locally free of bounded rank. Since the ranks of E and F are bounded, one may decompose T into a finite "disjoint sum" of opens on which E and F have constant rank. By using the compatibility of $K^\bullet$ with finite direct sums, and by writing the relations V 2.4.1 in the form V 2.4.2, one is reduced to the case of A-modules E, F locally free of finite constant rank.

If E has rank n, there is a torsor P under $\mathrm{GL}(n, A)$ such that

$$E = P \times_{\mathrm{GL}(n,A)} A^n,$$

where A^n is considered as a $\mathrm{GL}(n, A)$-module by the standard representation. One obtains a λ-homomorphism from $R_{\mathbb{Z}}(\mathrm{GL}(n))$ to K by sending a representation of $\mathrm{GL}(n)$ in a free $\mathbb{Z}$-module of rank p to the class of the locally free A-module $P \times_{\mathrm{GL}(n,A)} A^p$. The class of E is the image of the standard representation. Thus any polynomial relation between the $\lambda^i(E)$ follows from the corresponding relation for the standard representation in $R_{\mathbb{Z}}(\mathrm{GL}(n))$.

The same argument, with two locally free A-modules E, F of ranks n, p, uses a torsor under $\mathrm{GL}(n, A) \times \mathrm{GL}(p, A)$ and reduces the mixed relations between the $\lambda^i(E)$ and the $\lambda^j(F)$ to the representation ring $R_{\mathbb{Z}}(\mathrm{GL}(n) \times \mathrm{GL}(p))$. It remains only to know that, for every group scheme G over $\operatorname{Spec} \mathbb{Z}$, finite product of linear group schemes, the pre-λ-ring $R_{\mathbb{Z}}(G)$ is a λ-ring. For this one may refer to [5], where $R_{\mathbb{Z}}(G)$ is computed explicitly. Alternatively one can reason as in 3.2, using 1.14.

4. Computation of $K^\bullet$ of a flag bundle

4.1

Let S be a scheme and let E be a locally free $\mathcal{O}_S$-module of rank n. Let

$$\pi = (p_1, \dots, p_k)$$

be a sequence of integers ≥ 1 with

$$\sum_{i=1}^{k} p_i = n.$$

The flag functor of type π of E is the functor, on the category of S-schemes, which to an S-scheme $X \xrightarrow{f} S$ assigns the set of chains of quotients of $f^*(E)$

$$(L_{k-1}, \dots, L_1),$$

where L_i is a quotient of L_{i+1}, locally free of rank $p_1 + \cdots + p_i$.

This functor is representable by an S-scheme $D_\pi(E)$, endowed with a canonical chain of quotients of the inverse image of E, locally free of ranks

$$p_1, \quad p_1 + p_2, \quad \dots.$$

If $\pi = (1, n-1)$, then $D_\pi(E) = \mathbf{P}(E)$. If $\pi = (r, n-r)$, then $D_\pi(E) = G_{r,n}(E)$, the Grassmannian of type $(r, n-r)$ of E. If all p_i are equal to 1, one obtains the usual flag bundle of E, denoted $D(E)$.

Lemma 4.2. Let S be a scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, and let

$$\pi = (p_1, \dots, p_i, \dots, p_k), \qquad p_1 + \cdots + p_k = n.$$

Let $p', p'' \geq 1$ with $p' + p'' = p_i$, and put

$$\pi' = (p_1, \dots, p_{i-1}, p', p'', p_{i+1}, \dots, p_k), \qquad \pi'' = (p', p'').$$

Let $L_1, \dots, L_{k-1}$ be the canonical quotients of the inverse image of E on $D_\pi(E)$, and let F be the kernel of $L_i \to L_{i-1}$. Then there is a canonical S-isomorphism

$$D_{\pi'}(E) \simeq D_{\pi''}(F).$$

The proof is immediate from the functorial definition of flag schemes and is left to the reader.

Lemma 4.3. Let A be a commutative ring with unit, let $n > 0$, let $(c_i)_{1 \leq i \leq n}$ be a family of elements of A, and let C be the A-algebra generated by $\xi_1, \dots, \xi_n$ subject to the relations

$$\sigma_i(\xi_1, \dots, \xi_n) = c_i, \qquad i = 1, \dots, n,$$

where the σ_i are the elementary symmetric functions. Let $p_1, \dots, p_k$ be integers ≥ 1 with sum n, and, for $1 \leq j \leq k$, put

$$\xi^{(j)} = (\xi_1^{(j)}, \dots, \xi_{p_j}^{(j)}), \qquad \xi_i^{(j)} = \xi_{\sum_{\alpha<j} p_\alpha + i}.$$

Set

$$c_i^{(j)} = \sigma_i(\xi_1^{(j)}, \dots, \xi_{p_j}^{(j)}), \qquad 1 \leq j \leq k, \quad 1 \leq i \leq p_j,$$

and let B be the subalgebra of C generated by the $c_i^{(j)}$. Then:

i) The elements

$$(\xi^{(1)})^{\alpha(1)} \cdots (\xi^{(k)})^{\alpha(k)}, \qquad 0 \leq \alpha_i^{(j)} \leq p_j - i,$$

form a basis of C as a B-module.

ii) The generators $c_i^{(j)}$ of the A-algebra B satisfy the n isobaric relations deduced from

$$\sum_{i=1}^{n} c_i = \sum_{j=1}^{k} \sum_{i=1}^{p_j} c_i^{(j)}$$

(the c_i and $c_i^{(j)}$ being assigned weight i), and these n relations generate the ideal of relations among the $c_i^{(j)}$.

Corollary 4.4. The elements of C of the form

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \leq \alpha_i \leq n - i,$$

form a basis of C as an A-module.

For the proof of the lemma and of the corollary, we refer to [2], exposé 4, page 4.19, from which they are borrowed.

4.5

Let S be quasi-compact, let E be a locally free $\mathcal{O}_S$-module of rank n, and let $\pi = (p_1, \ldots, p_k)$ be a sequence of integers ≥ 1 with sum n. Put $X = D_\pi(E)$, and let X be endowed with its canonical chain of quotients L_i. Put $L_0 = 0$, $L_k = f^*(E)$, where $f : X \to S$ is the structural morphism. For $1 \leq j \leq k$, let F_j denote the kernel of $L_j \to L_{j-1}$. Let $K^\bullet(X)$ be the Grothendieck group of locally free sheaves of finite type on X, respectively of perfect complexes. Denote by F_j the class of F_j, and by $\lambda_i^{(j)}$ the class of $\lambda^i(F_j)$. In the case where $K^\bullet(X)$ is defined in terms of perfect complexes, the series $\lambda_t(E)$, $\lambda_t(F_j)$ are still given their evident meaning, where E is the class of E. Finally, note that

$$\sum_{j=1}^{k} F_j = E.$$

Theorem 4.6. With the hypotheses and notation of 4.5, the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $\lambda_i^{(j)}$. The ideal of relations among the $\lambda_i^{(j)}$ is generated by the relations deduced from

$$\lambda_t(E) = \prod_{j=1}^{k} \lambda_t(F_j).$$

Equivalently, $K^\bullet(X)$ is the $K^\bullet(S)$-λ-algebra generated by $F_1, \ldots, F_k$, subject to the relations (cf. V 4.6)

$$\text{i)}\quad \lambda^p(F_i) = 0 \quad \text{for } p > p_i, \qquad \text{ii)}\quad \sum_i F_i = E.$$

We begin with the case where all p_i are equal to 1. The theorem then says the following.

Corollary 4.7. Let S be a quasi-compact scheme, let E be a locally free $\mathcal{O}_S$-module of rank n, let X be the flag bundle of E, and let $\xi_1, \ldots, \xi_n$ be the classes of the invertible sheaves $F_1, \ldots, F_n$ (notation 3.5). Then $K^\bullet(X)$ is the $K^\bullet(S)$-algebra generated by the ξ_i, subject to the relations

$$\sigma_i(\xi_1, \ldots, \xi_n) = \lambda^i(E).$$

The proof is by induction on n. For $n = 1$ the assertion is trivial. Assume it known for $n - 1$, and consider $\mathbf{P}(E)$ with the exact sequence

$$0 \longrightarrow F \longrightarrow g^*(E) \longrightarrow \mathcal{O}_{\mathbf{P}(E)}(1) \longrightarrow 0,$$

where g is the structural morphism of $\mathbf{P}(E)$. By Lemma 4.2, X is identified with the usual flag bundle $D(F)$ over $\mathbf{P}(E)$, and the canonical sheaves correspond under this identification. The diagram

$$\begin{tikzcd} D(E) = D(F) \arrow[r, "h"] \arrow[dr, "f"'] & \mathbf{P}(E) \arrow[d, "g"] \\ & S \end{tikzcd}$$

shows that $K^\bullet(S) \to K^\bullet(X)$ factors through $K^\bullet(\mathbf{P}(E))$. By the induction hypothesis applied to $D(F)$, and by Corollary 3.4, $K^\bullet(X)$ has a $K^\bullet(\mathbf{P}(E))$-basis consisting of the elements

$$\xi_2^{\alpha_2} \cdots \xi_n^{\alpha_n}, \qquad 0 \leq \alpha_i \leq n-i.$$

Moreover, $K^\bullet(\mathbf{P}(E))$ has $K^\bullet(S)$-basis

$$1, \xi_1, \ldots, \xi_1^{n-1}.$$

Thus $K^\bullet(X)$ has, as a $K^\bullet(S)$-basis, the elements

$$\xi_1^{\alpha_1} \cdots \xi_{n-1}^{\alpha_{n-1}}, \qquad 0 \leq \alpha_i \leq n-i.$$

Additivity of λ_t gives

$$\prod_i (1+\xi_i t) = \lambda_t(E),$$

and hence the relations of the corollary. Conversely, if C is the $K^\bullet(S)$-algebra generated by ξ_i subject to the relations

$$\sigma_i(\xi_1, \ldots, \xi_n) = \lambda^i(E),$$

then there is a surjective homomorphism $C \to K^\bullet(X)$. By 4.4 this homomorphism sends a basis of C to a basis of $K^\bullet(X)$, hence is an isomorphism.

We now treat the general flag bundle of type π. The usual flag bundle $D(E)$ is obtained from $D_\pi(E)$ by successively constructing the flag bundles of the F_j's (4.2). It follows that $K^\bullet(D(E))$, considered as a module over $K^\bullet(X)$, has a basis formed by the elements considered in 4.3.i), the $\xi_i^{(j)}$ being the classes of the sheaves analogous to the F_j's which occur in the splitting of the inverse images of the F_j's on $D(E)$. Thus $K^\bullet(X)$ is a subring of $K^\bullet(D(E))$, and additivity of λ_t gives

$$\lambda_i^{(j)} = \sigma_i(\xi_1^{(j)}, \ldots, \xi_{p_j}^{(j)}).$$

Let B be the subalgebra generated by these elements. By 4.3.i), the same family is a basis of $K^\bullet(D(E))$ both as a B-module and as a $K^\bullet(X)$-module; hence $K^\bullet(X) = B$, which proves that $K^\bullet(X)$ is generated by the $\lambda_i^{(j)}$. The ideal of relations is then given by 4.3.ii), in view of the $K^\bullet(S)$-module structure of $K^\bullet(D(E))$, determined by 4.7, and the fact that the $\xi_i^{(j)}$ are the classes of the sheaves occurring in the splitting of the inverse image of E on $D(E)$. This proves the first form of 4.6.

To deduce the second form, observe that the elements F_i of the $K^\bullet(S)$-λ-algebra $K^\bullet(X)$ satisfy relations i) and ii). If A is the $K^\bullet(S)$-λ-algebra generated by elements F'_i satisfying i) and ii), there is a $K^\bullet(S)$-λ-homomorphism $A \to K^\bullet(X)$ sending F'_i to F_i (V 4.7). By 4.6 this homomorphism is surjective. Since the λ-ideal generated by relations i) and ii) contains the ordinary ideal generated by the relations of the first form of 4.6, the homomorphism is also injective, hence an isomorphism.

Corollary 4.8. With the hypotheses and notation of 4.5, put

$$c_i^{(j)} = \gamma^i(F_j - p_j), \qquad 1 \leq i \leq p_j.$$

Then the $K^\bullet(S)$-algebra $K^\bullet(X)$ is generated by the $c_i^{(j)}$, and the ideal of relations among them is generated by the relations deduced from

$$\gamma_t(E-n) = \prod_{j=1}^{k} \gamma_t(F_j - p_j).$$

This is merely another statement of 4.6: one can express the $c_i^{(j)}$ as functions of the $\lambda_i^{(j)}$, and conversely. Moreover, $\lambda_t(E) = \prod \lambda_t(F_j)$ is equivalent to

$$\gamma_t(E-n) = \prod_{j} \gamma_t(F_j - p_j).$$

4.9

Let S be a scheme. Denote by $G_S^{n,r}$ the Grassmannian of type $(n, r-n)$ of $(\mathcal{O}_S)^r$, with $r \geq n$. As r varies, the $G_S^{n,r}$'s form an inductive system, the morphism

$$G_S^{n,r} \longrightarrow G_S^{n,r+1}$$

being defined by considering $(\mathcal{O}_S^n)^r$ as a quotient of $(\mathcal{O}_S^n)^{r+1}$ by the $(r+1)$-st factor. In this system the canonical quotients correspond by inverse image. The inductive system thus defined will be denoted $G_S^{n,\infty}$. Its $K^\bullet$ is defined by

$$K^\bullet(G_S^{n,\infty}) = \varprojlim_r K^\bullet(G_S^{n,r}). \tag{4.9.1}$$

Let F_1, F_2 be the canonical sheaves on $G_S^{n,r}$ (4.5), and denote also by F_1, F_2 their classes in $K^\bullet(G_S^{n,r})$. Put

$$c_i = \gamma^i(F_1 - n).$$

When r varies, the c_i's are compatible with the transition morphisms and hence define elements, again denoted c_i, of $K^\bullet(G_S^{n,\infty})$.

Proposition 4.10. Let S be quasi-compact. With the notation of 4.9, one has an isomorphism

$$K^\bullet(G_S^{n,\infty}) \simeq K^\bullet(S)[[c_1, \ldots, c_n]].$$

Indeed, by 4.8, $K^\bullet(G_S^{n,r})$ is the $K^\bullet(S)$-algebra generated by

$$c_i = \gamma^i(F_1 - n), \quad 1 \leq i \leq n,$$

and by the $\gamma^j(F_2 - r + n)$, $1 \leq j \leq r - n$, subject to the relations deduced from

$$\gamma_t(F_1 - n)\gamma_t(F_2 - r + n) = \gamma_t(r - r) = 1,$$

that is,

$$\gamma_t(F_2 - r + n) = \frac{1}{\gamma_t(F_1 - n)}.$$

This relation expresses the $\gamma^j(F_2 - r + n)$ in terms of the c_i's, and gives a series of isobaric relations among the c_i's (with c_i of weight i) by requiring that the coefficients of $1/\gamma_t(F_1 - n)$ vanish in degrees $> r - n$. These relations generate the ideal of relations among the c_i's. Thus

$$K^\bullet(G_S^{n,r}) = K^\bullet(S)[c_1, \ldots, c_n]/I_r,$$

where I_r is a homogeneous ideal of degree $> r - n$. It follows from (4.9.1) that $K^\bullet(G_S^{n,\infty})$ is the separated completion of $K^\bullet(S)[c_1, \ldots, c_n]$ for the topology defined by the I_r's. This topology is finer than the topology defined by weights; we shall show that it is equivalent to it, which proves the proposition.

Put $A = K^\bullet(S)[c_1, \dots, c_n]/I_r$, and let C be the A-algebra generated by $x_1, \dots, x_n$, subject to

$$\sigma_j(x_1, \dots, x_n) = c_j.$$

By 4.4, the canonical homomorphism $A \to C$ is injective. In C one has

$$\gamma_t(F_1 - n) = 1 + c_1 t + \cdots + c_n t^n = \prod_{i=1}^{n}(1 + x_i t).$$

Since $1/\gamma_t(F_1 - n)$ is a polynomial, the same is true of $1/(1 + x_i t)$ for every i. Hence the x_i's are nilpotent, and there is an integer N such that every polynomial of degree $> N$ in the x_i's, with coefficients in $K^\bullet(S)$, is zero. Consequently every polynomial in A of weight $> N$ is zero, i.e. every polynomial in $K^\bullet(S)[c_1, \dots, c_n]$ of weight $> N$ belongs to I_r. Thus the two topologies considered are equivalent, and the proof is complete.

5. Applications to projective bundles; study of f_*

5.1

In this section, S is a quasi-compact scheme, E is a locally free $\mathcal{O}_S$-module of rank $r+1$, and X is the projective bundle associated to E. On X there is the canonical exact sequence

$$0 \longrightarrow F \longrightarrow f^*(E) \longrightarrow \mathcal{O}_X(1) \longrightarrow 0,$$

where $f : X \to S$ is the structural morphism.

The $K^\bullet$'s considered here are those defined using locally free sheaves of finite type. We denote by E, F, ξ the classes of $E, F, \mathcal{O}_X(1)$. Put

$$x = 1 - \xi^{-1}, \qquad x' = \xi - 1 = \xi x,$$

and denote by $\bar{x}, \bar{x}'$ the classes of x, x' in $\mathrm{Gr}^1 K^\bullet(X)$. Since $\xi - 1$ and x are in $\mathrm{Fil}^1 K^\bullet(X)$, one has

$$(\xi - 1)x \in \mathrm{Fil}^2 K^\bullet(X), \quad \text{i.e.} \quad x \equiv \bar{\xi} x = x' \pmod{\mathrm{Fil}^2 K^\bullet(X)},$$

or equivalently $\bar{x}' = -\bar{x}$, according to the sign convention for the associated graded used here.

5.2

If G is a locally free finite-type $\mathcal{O}_X$-module, the $R^i f_*(G)$ are not, in general, locally free of finite type on S. Hence one cannot define a homomorphism $f_* : K^\bullet(X) \to K^\bullet(S)$ as in the case of $K^\bullet$'s defined from perfect complexes. Theorem 1.1 avoids this difficulty as follows: if $y \in K^\bullet(X)$, one can write uniquely

$$y = a_0 + a_1 \xi^{-1} + \cdots + a_r \xi^{-r}, \qquad a_i \in K^\bullet(S), \tag{5.2.1}$$

and one puts

$$f_*(y) = a_0. \tag{5.2.2}$$

It is clear that the homomorphism so defined is additive and satisfies the projection formula, i.e. is $K^\bullet(S)$-linear. The following relations are verified:

$$f_*(1) = 1, \qquad f_*(\xi^n) = 0 \ (-r \le n < 0), \qquad f_*(\xi^n) = \sigma_n \quad (1 \le n). \tag{5.2.3}$$

To check the last group of relations, one proceeds by induction on n. If $f_*(\xi^p) = \sigma_p$ for $p \le n-1$, relation (1.9.1), multiplied by a suitable power of ξ, gives

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) \xi^{n-i} = 0.$$

Applying f_*, one obtains

$$\sum_{i=0}^{r+1} (-1)^i \lambda^i(E) f_*(\xi^{n-i}) = 0,$$

and the result follows at once from relation (1.11.3) and the induction hypothesis.

Proposition 5.3. With the notation and hypotheses of 5.1, for every $k \ge 0$ one has a direct-sum decomposition of $K^\bullet(S)$-modules

$$\operatorname{Fil}^k K^\bullet(X) = \operatorname{Fil}^k K^\bullet(S) + \operatorname{Fil}^{k-1} K^\bullet(S) x + \cdots + \operatorname{Fil}^{k-r} K^\bullet(S) x^r,$$

where the filtration considered is the augmented λ-algebra filtration defined in V 3.9.1 and 3.10.

First note that x has zero augmentation. Moreover

$$\gamma_t(x) = \lambda_{t/(1-t)}(1 - \xi^{-1}) = \left(1 + \frac{t}{1-t}\right)\left(1 + \frac{\xi^{-1} t}{1-t}\right)^{-1} = \frac{1}{1 - xt}.$$

Hence $\gamma^i(x) = x^i$ for all i. Thus $K^\bullet(X)$ is a quotient of the $K^\bullet(S)$-λ-algebra generated by a generator X subject to the relations $\gamma^i(X) = X^i$ for all i, and the quotient ideal is generated by relation (1.13.1). One may therefore apply V 4.12 and V 4.13(ii), and one is reduced to showing that for every $n \ge 0$

$$x^n \in \operatorname{Fil}^n K^\bullet(S) + \operatorname{Fil}^{n-1} K^\bullet(S) x + \cdots + \operatorname{Fil}^{n-r} K^\bullet(S) x^r.$$

This is clear for $n \le r$, and the rest follows by induction. Relation (1.13.1) gives

$$x^n = -\sum_{i=0}^{r} Y^{r+1-i}(E - r - 1)\, x^{n-r-1+i}, \qquad n \ge r+1,$$

and the induction hypothesis applied to the powers $x^{n-r-1+i}$ gives the assertion. The sum is direct because $1, \dots, x^r$ are independent over $K^\bullet(S)$.

Corollary 5.4. The filtration induced by $K^\bullet(X)$ on $K^\bullet(S)$ is identical with the filtration of $K^\bullet(S)$.

Corollary 5.5. Let S be quasi-compact, let $E_1, \dots, E_n$ be locally free $\mathcal{O}_S$-modules of finite type, let $D(E_i)$ be the flag scheme of E_i, and put $X = D(E_1) \times_S \cdots \times_S D(E_n)$. Then the canonical homomorphism

$$\operatorname{Gr} K^\bullet(S) \longrightarrow \operatorname{Gr} K^\bullet(X)$$

is injective.

The reasoning of 3.1 reduces this to the case where X is a projective bundle over S, and then the assertion follows at once from 5.4.

Corollary 5.6. With the notation and hypotheses of 5.1, $\operatorname{Gr} K^\bullet(X)$ is a free $\operatorname{Gr} K^\bullet(S)$-module with basis $1, \bar{x}, \dots, \bar{x}^r$. For every k,

$$\operatorname{Gr}^k K^\bullet(X) = \operatorname{Gr}^k K^\bullet(S) \oplus \operatorname{Gr}^{k-1} K^\bullet(S) \bar{x} \oplus \cdots \oplus \operatorname{Gr}^{k-r} K^\bullet(S) \bar{x}^r.$$

Moreover, the elements $1, \bar{x}, \ldots, \bar{x}^{r+1}$ are linked by

$$\sum_{i=0}^{r+1} c^i(E)\bar{x}^{\,r+1-i} = 0,$$

where $c^k(E)$ is the k-th Chern class of E (V 6.7). This is an immediate consequence of 1.1, 5.3, and (1.13.1).

Corollary 5.7. With the notation and hypotheses of 4.5, the $\operatorname{Gr} K^\bullet(S)$-algebra $\operatorname{Gr} K^\bullet(X)$ is generated by the $\bar{c}_i^{(j)}$, where $\bar{c}_i^{(j)}$ is the class in $\operatorname{Gr}^i K^\bullet(X)$ of $\gamma^i(F_j - p_j)$. The ideal of relations among the $\bar{c}_i^{(j)}$ is generated by the relations deduced from

$$c(E-n) = \prod_{j=1}^{k} c(F_j - p_j),$$

where c is the total Chern class (V 6.7). This result is proved by copying the proof of 4.6 and using 5.6 (compare [2], Exp. 4).

Corollary 5.8. With the notation and hypotheses of 5.1, for every k

$$f_*(\operatorname{Fil}^k K^\bullet(X)) \subset \operatorname{Fil}^{k-r} K^\bullet(S).$$

This is an immediate consequence of 5.3 and the projection formula.

Proposition 5.9. With the notation and hypotheses of 5.1, for every $i \geq 1$

$$f_*(\lambda^i(F)) = 0 \qquad \text{and hence} \qquad f_*(\gamma^i(F)) = 0.$$

In particular, putting

$$\lambda_{-1}(F) = \sum_{i\geq 0} (-1)^i \lambda^i(F),$$

one obtains

$$f_*(\lambda_{-1}(F)) = 1.$$

Define a map

$$\overline{f}_* : 1 + K^\bullet(X)[[t]]^+ \longrightarrow 1 + K^\bullet(S)[[t]]^+$$

by

$$\overline{f}_*\left(\sum_{i\geq 0} a_i t^i\right) = \sum_{i\geq 0} f_*(a_i) t^i.$$

It is not in general a group homomorphism; nevertheless, for $\varphi \in 1 + K^\bullet(X)[[t]]^+$ and $\psi \in 1 + K^\bullet(S)[[t]]^+$, one has

$$\overline{f}_*(\varphi\, g(f^*(\psi))) = \psi\, \overline{f}_*(\varphi),$$

coefficient by coefficient, by the projection formula. From $E = F + \xi$ one deduces

$$\overline{f}_*(\lambda_t(F)) = \overline{f}_*(\lambda_t(E)\lambda_t(-\xi)) = \lambda_t(E)\, \overline{f}_*(\lambda_t(-\xi)).$$

But

$$\overline{f}_*(\lambda_t(-\xi)) = \sum_{i\geq 0} (-1)^i f_*(\xi^i) t^i = \sigma_{-t}(E)$$

by (4.2.3). Using relation (1.12.1), one finally obtains

$$\overline{f}_*(\lambda_t(F)) = 1,$$

and the desired relations follow.

Proposition 5.10. With the notation and hypotheses of 5.1, let $z \in K^\bullet(X)$. The condition $z(1-\xi) = 0$ is necessary and sufficient for the existence of $y \in K^\bullet(S)$ such that

$$\text{i)} \quad z = y\,\lambda_{-1}(F), \qquad \text{ii)} \quad y\,\lambda_{-1}(E) = 0.$$

Since $E = F + \xi$, one has

$$\lambda_{-1}(E) = \lambda_{-1}(F)\lambda_{-1}(\xi) = \lambda_{-1}(F)(1-\xi),$$

so the condition is sufficient.

Conversely, suppose $z(1-\xi) = 0$, or $zx' = 0$. By Remark 1.13 one can write uniquely

$$z = \sum_{i=0}^{r} a_i (x')^i, \qquad a_i \in K^\bullet(S).$$

Moreover, x' satisfies an equation of dependence

$$\sum_{i=0}^{r} c_i (x')^i + (x')^{r+1} = 0, \qquad c_i = (-1)^{r+i+1} \Upsilon^{r+1-i}(E - r - 1).$$

Since $zx' = 0$, one obtains

$$a_r (x')^{r+1} = -\sum_{i=1}^{r} a_{i-1}(x')^i = -\sum_{i=0}^{r} a_r c_i (x')^i.$$

Using the independence of $1, \ldots, (x')^r$, one finds

$$a_r c_0 = 0, \qquad a_i = a_r c_{i+1} \quad (0 \le i < r).$$

Hence

$$z = a_r \left(\sum_{i=0}^{r-1} c_{i+1}(x')^i + (x')^r \right).$$

The expression in parentheses is $\lambda_{-1}(F)$; therefore, putting $y = a_r$, one gets $z = y\lambda_{-1}(F)$. Moreover $yc_0 = 0$, and $c_0 = \lambda_{-1}(E)$, hence $y\lambda_{-1}(E) = 0$.

Proposition 5.11. With the notation and hypotheses of 5.1, let $z \in \operatorname{Gr} K^\bullet(X)$. The condition $z\bar{x}' = 0$ is necessary and sufficient for the existence of $y \in \operatorname{Gr} K^\bullet(S)$ such that

$$\text{i)} \quad z = y\,c^r(F), \qquad \text{ii)} \quad y\,c^{r+1}(E) = 0.$$

The proof is identical with that of 5.10, after passing to the associated graded rings and using 5.6.

6. Study of the filtration of $K^\bullet(X)$ when X has an ample sheaf

In this section X is a scheme having an ample sheaf L. Recall that in this case the $K^\bullet$'s defined in terms of locally free sheaves of finite type and those defined in terms of perfect complexes are equal (IV 2.9). Let

$$\varepsilon : K^\bullet(X) \longrightarrow H^0(X, \mathbb{Z})$$

be the augmentation homomorphism.

Proposition 6.1. Let X be a quasi-compact scheme with an ample sheaf L. Then $\operatorname{Fil}^1 K^\bullet(X)$ is a nilideal.

Since $\operatorname{Fil}^1 K^\bullet(X)$ is generated by the elements $E - \varepsilon(E)$, where E is the class of a locally free finite-type $\mathcal{O}_X$-module E, it is enough to show that such an element is nilpotent. Quasi-compactness and the compatibility of $K^\bullet$ with finite direct sums of opens reduce the proof to the case where E has constant rank. Passing to the flag bundle of E, which gives an injection on $K^\bullet$ by 3.1 and preserves the existence of an ample sheaf, reduces further to the case where E is invertible.

Let E^* be the dual of E. For n large enough,

$$E^*(n) = E^* \otimes_{\mathcal{O}_X} L^{\otimes n}$$

is generated by global sections. Hence there is an integer m and a surjective homomorphism

$$(\mathcal{O}_X)^m \longrightarrow E^*(n),$$

or, after tensoring by $E(-n)$, a surjection

$$(E(-n))^m \longrightarrow \mathcal{O}_X.$$

The kernel is locally free of rank $m-1$. Denoting by L the class of the ample sheaf, one obtains

$$\lambda^m(mEL^{-n} - 1) = 0.$$

But

$$\lambda^m(mEL^{-n} - 1) = (-1)^m \lambda^m_{-1}(mEL^{-n}) = (-1)^m \lambda_{-1}(EL^{-n})^m = (-1)^m(1 - EL^{-n})^m.$$

Thus, for every invertible sheaf E, the element $1 - EL^{-n}$ is nilpotent for n large, or equivalently its product $L^n - E$ by L^n is nilpotent. Finally

$$E - 1 = (L^n - 1) - (L^n - E),$$

and the two terms on the right are nilpotent for n large; hence $E - 1$ is nilpotent.

Corollary 6.2. With the hypotheses of 6.1, let $R \subset K^\bullet(X)$ be a sub-λ-ring generated by the classes of finitely many sheaves. Then the λ-ring filtration of R is discrete, and $R \cap \operatorname{Fil}^1 K^\bullet(X)$ is nilpotent. Moreover, R is a finite-type module over $R \cap H^0(X, \mathbb{Z})$ (V 3.9.1).

Since X is quasi-compact, one can find a finite covering of X by disjoint opens on each of which the sheaves whose classes generate R all have constant rank. The standard argument reduces to the case where these sheaves have constant rank on X; then R is augmented over $\mathbb{Z}$. If E_i, $i = 1, \ldots, k$, are the sheaves whose classes generate R, write E_i for their classes and $x_i = E_i - \varepsilon(E_i)$. Then R is a quotient of the λ-ring generated by the x_i, subject to relations

$$\gamma^j(x_i) = 0 \quad (j > r_i = \varepsilon(E_i)),$$

namely of

$$\mathbb{Z}\left[x_i, \gamma^j(x_i)\right]_{1 \le i \le k,\, 2 \le j \le r_i}.$$

By Proposition 6.1 the x_i's and the $\gamma^j(x_i)$'s are nilpotent. Thus the augmentation ideal of R is nilpotent; it is precisely $R \cap \operatorname{Fil}^1 K^\bullet(X)$. Moreover every element of Fil^n_R is represented by a polynomial without components of weight $< n$ in the $x_i, \gamma^j(x_i)$. If m is the largest integer for which some $\gamma^{m-1}(x_i) \neq 0$, and if p is an integer such that all p-th powers of the $x_i, \gamma^j(x_i)$ vanish, then $\operatorname{Fil}^n_R = 0$ for $n > kp(1 + \cdots + m)$. Hence the filtration of R is discrete. Since only finitely many monomials in the $x_i, \gamma^j(x_i)$ survive, they form a system of generators of R as a $\mathbb{Z}$-module, completing the proof.

Corollary 6.3. With the hypotheses of 6.1, let A be a commutative ring. An element $y \in K^\bullet(X) \otimes_{\mathbb{Z}} A$ is invertible if and only if $(\varepsilon \otimes \mathrm{id}_A)(y)$ is invertible. In particular, if E is a locally free finite-type sheaf on X, everywhere non-zero, then its class in $K^\bullet(X)_{\mathbb{Q}}$ is invertible.

The augmentation ideal of $K^\bullet(X) \otimes_{\mathbb{Z}} A$ is $\mathrm{Fil}^1 K^\bullet(X) \otimes_{\mathbb{Z}} A$; by 6.1 it is a nilideal. Thus an element is invertible precisely when its image under the augmentation is invertible.

Corollary 6.4. Let X be quasi-compact with an ample sheaf, and let $f : X' \to X$ be a proper and perfect morphism. Assume moreover one of the following two conditions:

$$\text{i) } f \text{ is projective,} \qquad \text{ii) } X \text{ is noetherian,}$$

so that the "finiteness theorem" holds (III 2.2). Let A be a ring and consider the direct-image homomorphism (IV 2.11)

$$f_* : K^\bullet(X') \otimes_{\mathbb{Z}} A \longrightarrow K^\bullet(X) \otimes_{\mathbb{Z}} A.$$

Assume that there exists $x' \in K^\bullet(X') \otimes_{\mathbb{Z}} A$ such that

$$\varepsilon(f_*(x')) \in H^0(X, A)$$

is invertible. Then

$$f^* : K^\bullet(X) \otimes_{\mathbb{Z}} A \longrightarrow K^\bullet(X') \otimes_{\mathbb{Z}} A$$

is injective.

Indeed, the projection formula gives

$$f_*(f^*(x)x') = x\, f_*(x').$$

The element $f_*(x')$ has invertible augmentation and is therefore invertible by 6.3. Hence the composite

$$x \longmapsto f^*(x) \longmapsto f^*(x)x' \longmapsto f_*(f^*(x)x')$$

is injective, and so f^* is injective.

6.5

Let X be a noetherian scheme of dimension d with an ample sheaf L. We endow $K^\bullet(X)$ with the following filtration (cf. [1]): an element x belongs to $K^\bullet(X)_i$ if and only if, for every closed subset Y of X, there exists a perfect complex $M^\bullet(Y)$ such that

$$\text{i)} \quad \mathrm{cl}(M^\bullet(Y)) = x,$$

$$\text{ii)} \quad \mathrm{codim}(\mathrm{Supp}(H^*(M^\bullet(Y))) \cap Y,\ Y) \geq i, \qquad H^*(M^\bullet(Y)) = \bigoplus_{i \geq 0} H^i(M^\bullet(Y)).$$

The subgroups $K^\bullet(X)_i$ so defined are additive subgroups of $K^\bullet(X)$. Indeed, if $x, x' \in K^\bullet(X)_i$, and if Y is a closed subset of X, with corresponding complexes $M^\bullet$ and $M'^\bullet$, then $x + x'$ is the class of $M^\bullet \oplus M'^\bullet$, and

$$\mathrm{Supp}(H^*(M^\bullet \oplus M'^\bullet)) = \mathrm{Supp}(H^*(M^\bullet)) \cup \mathrm{Supp}(H^*(M'^\bullet)).$$

Since the codimension of a finite union of closed subsets is the lower bound of the codimensions of the closed subsets, one has $x + x' \in K^\bullet(X)_i$.

Proposition 6.6. The filtration 6.5 of $K^\bullet(X)$ is a ring filtration, and

$$K^\bullet(X)_{d+1} = 0.$$

Let $x \in K^\bullet(X)_i$, $x' \in K^\bullet(X)_j$, and let Y be a closed subset of X. There exist perfect complexes $M^\bullet$, $M'^\bullet$ representing x, x', and such that

$$\operatorname{codim}(\operatorname{Supp}(H^*(M^\bullet)) \cap Y, Y) \geq i,$$

$$\operatorname{codim}(\operatorname{Supp}(H^*(M'^\bullet)) \cap \operatorname{Supp}(H^*(M^\bullet)) \cap Y,\ \operatorname{Supp}(H^*(M^\bullet)) \cap Y) \geq j.$$

Then xx' is represented by $M^\bullet \otimes M'^\bullet$, and

$$\operatorname{codim}(\operatorname{Supp}(H^*(M^\bullet \otimes M'^\bullet)) \cap Y, Y) \geq i + j$$

by EGA 0_{IV}, 14.2.2.3, since

$$\operatorname{Supp}(H^*(M^\bullet \otimes M'^\bullet)) \subset \operatorname{Supp}(H^*(M^\bullet)) \cap \operatorname{Supp}(H^*(M'^\bullet)).$$

This proves that the filtration is multiplicative. To show $K^\bullet(X)_{d+1} = 0$, take $x \in K^\bullet(X)_{d+1}$ and $Y = X$. There is a perfect complex $M^\bullet$ with $x = \operatorname{cl}(M^\bullet)$ and

$$\operatorname{codim}_X(\operatorname{Supp}(H^*(M^\bullet))) \geq d + 1.$$

Since X has dimension d, this forces $\operatorname{Supp}(H^*(M^\bullet)) = \varnothing$, so $M^\bullet$ is acyclic and $x = 0$.

Lemma 6.7. Let $X = \operatorname{Spec}(A)$ be an affine noetherian scheme, let $x_1, \dots, x_k$ be finitely many points of X, and let E be an invertible $\mathcal{O}_X$-module. Then there exists a section $s \in \Gamma(X, E)$ such that, for every i, s_{x_i} is a basis of E_{x_i}.

Let $M = \Gamma(X, E)$. For every i, choose a closed point $x'_i \in \overline{\{x_i\}}$. The x'_i's correspond to maximal ideals $\mathfrak{m}_i$ of A, which may be assumed all distinct after changing the index set so that each $\mathfrak{m}_i$ occurs once. Since E is invertible, $M_{\mathfrak{m}_i} \simeq A_{\mathfrak{m}_i}$, hence $M/\mathfrak{m}_i M \simeq A/\mathfrak{m}_i$. Thus

$$\prod_i M/\mathfrak{m}_i M \simeq \prod_i A/\mathfrak{m}_i, \qquad A/\prod_i \mathfrak{m}_i \simeq M/\left(\prod_i \mathfrak{m}_i\right) M.$$

Lifting the latter isomorphism gives a homomorphism $A \to M$ whose localization $A_{\mathfrak{m}_i} \to M_{\mathfrak{m}_i}$ is an isomorphism modulo $\mathfrak{m}_i$, hence an isomorphism by Nakayama. The image of 1 gives a section s of E which is a basis at the x'_i, and therefore at the x_i.

Lemma 6.8. Let X be a noetherian scheme with an ample sheaf L, let E be an invertible $\mathcal{O}_X$-module, and let $x_1, \dots, x_k$ be points of X. Then there is an integer n such that, for every $k \geq 1$, there exists a section

$$s \in \Gamma(X, E(kn))$$

which is a basis of $E(kn)_{x_i}$ for every i.

One can find an integer n_0 and a section $f \in \Gamma(X, L^{\otimes n_0})$ such that every x_i belongs to the affine open X_f (EGA II 4.5.4). On X_f, Lemma 6.7 gives a section $t \in \Gamma(X_f, E)$ satisfying the required conditions; moreover, for some integer n'_0, the section $f^{\otimes n'_0} t$ extends to all of X. Put $n = n_0 n'_0$. Then $f^{\otimes n'_0} \in \Gamma(X, L^{\otimes n})$; tensoring the section of $E(n)$ obtained above by powers of $f^{\otimes kn'}$ gives, for every $k \geq 0$, a section of $E((k+1)n)$. Since every x_i lies in X_f and t is a basis there, these sections give bases of $E(kn)_{x_i}$ for every $k \geq 1$.

Theorem 6.9. Let X be a noetherian scheme of dimension d with an ample sheaf L. Then

$$\operatorname{Fil}^{d+1} K^\bullet(X) = 0, \qquad \text{and a fortiori} \qquad (\operatorname{Fil}^1 K^\bullet(X))^{d+1} = 0.$$

The last relation is due to J.-P. Serre in the general case considered here.

First we show that, if E, E' are invertible sheaves on X, with classes E, E', then $E - E' \in K^\bullet(X)_1$. Let Y be a closed subset of X, let $Y_1, \ldots, Y_k$ be its irreducible components, and choose $x_i \in Y_i - \bigcup_{j \neq i} Y_j$. By Lemma 6.8 there are integers n, n' such that, for arbitrary $k, k' \geq 1$, there exist homomorphisms

$$L^{-kn} \xrightarrow{s_k} E, \qquad L^{-k'n'} \xrightarrow{s'_{k'}} E'$$

which are isomorphisms at the points x_i. Consider the complex

$$N^\bullet : \quad \cdots \longrightarrow 0 \longrightarrow L^{-nn'} \xrightarrow{(s'_n, 0)} E' \oplus L^{-nn'} \xrightarrow{0+s_n} E \longrightarrow 0 \longrightarrow \cdots .$$

The class of $N^\bullet$ is $E - E'$. Moreover $N^\bullet$ is acyclic at the x_i; hence $\operatorname{Supp}(H^*(N^\bullet)) \cap Y$ contains no irreducible component of Y, and so has codimension at least 1 in Y. This proves $E - E' \in K^\bullet(X)_1$.

Let $x \in \operatorname{Fil}^{d+1} K^\bullet(X)$. We shall prove that $x = 0$. One may suppose that x has the form

$$x = \gamma^{i_1}(E_1 - F_1) \cdots \gamma^{i_k}(E_k - F_k), \qquad i_1 + \cdots + i_k \geq d + 1,$$

where the E_i, F_i are classes of locally free finite-type sheaves of the same rank; after decomposing X into finitely many opens, one may assume these ranks are constant. Let X' be the product, over X, of the flag schemes of the E_i, F_i. In $K^\bullet(X')$, their images are sums of classes of invertible sheaves. If ξ, η are classes of invertible $\mathcal{O}_{X'}$-modules, then

$$\gamma_t(\xi - \eta) = \frac{\gamma_t(\xi - 1)}{\gamma_t(\eta - 1)} = \frac{1 + (\xi - 1)t}{1 + (\eta - 1)t},$$

and therefore

$$\gamma^i(\xi - \eta) = (-1)^{i-1}(\xi - \eta)(\eta - 1)^{i-1}.$$

Since X' is noetherian and has an ample sheaf, the preceding result and 6.6 give

$$\gamma^i(\xi - \eta) \in K^\bullet(X')_i.$$

Using the additivity of γ_t and 6.6, it follows that

$$f^*(\gamma^{i_j}(E_j - F_j)) \in K^\bullet(X')_{i_j},$$

and hence $f^*(x) \in K^\bullet(X')_{d+1}$, where $f : X' \to X$ is the canonical morphism.

We now deduce that $f^*(x) = 0$. Let r be the relative dimension of X' over X. There exists an element $x' \in K^\bullet(X')$, belonging to a basis of $K^\bullet(X')$ over $K^\bullet(X)$, of the form

$$x' = \prod_{i=0}^{r} (\xi_i - 1),$$

where the ξ_i's are classes of invertible sheaves. To prove this, the argument of 3.1 reduces one to the case of a projective bundle over X, where it follows immediately from 1.13. The element x' lies in $K^\bullet(X')_r$ by what precedes, hence

$$x' f^*(x) \in K^\bullet(X')_{d+r+1}.$$

But $\dim X' = d + r$; therefore 6.6 gives $x' f^*(x) = 0$. Since x' is part of a basis of $K^\bullet(X')$ over $K^\bullet(X)$, this implies $f^*(x) = 0$. Since f^* is injective by 3.1, one finally obtains $x = 0$.

Problem 6.10. Under the hypotheses of 6.9, is it true that for every k one has

$$\operatorname{Fil}^k K^\bullet(X) \subset K^\bullet(X)_k?$$

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